Hossein Mohebbi

hossein.mohebbi@uwaterloo.ca · Google Scholar · github.com/mhmohebbi · linkedin.com/in/hossein-mohebbi · hosseinmohebbi.com

RESEARCH INTERESTS

Reinforcement learning, causal inference and robust machine learning, with applications across tabular and generative settings. My current focus is computer-use agents and multi-agent systems.

EDUCATION

University of Waterloo, Waterloo, ON — 2023 – 2026
Master of Mathematics (MMath), Computer Science

- Thesis: *Counterfactual Data Augmentation for Regression*
- Advisor: Pascal Poupart

University of Waterloo, Waterloo, ON — 2018 – 2023
Bachelor of Computer Science (BCS), Honours, Co-operative Program

- Human–Computer Interaction Option
- Dean's Honours

PUBLICATIONS

Peer-reviewed conference papers

1. **Counterfactual Residual Data Augmentation for Regression.**
   Hossein Mohebbi, Oliver Schulte, Ke Li, Pascal Poupart.
   43rd International Conference on Machine Learning (ICML 2026), Seoul, South Korea.
   arXiv:2606.28460 · Project page · Code

Preprints and manuscripts under review

2. **Image-POSER: Reflective RL for Multi-Expert Image Generation and Editing.**
   Hossein Mohebbi[†], Mohammed Abdulrahman[†], Yanting Miao, Pascal Poupart, Suraj Kothawade.
   Under review. † Equal contribution.
   arXiv:2511.11780

Thesis

3. **Counterfactual Data Augmentation for Regression.**
   Hossein Mohebbi.
   MMath thesis, University of Waterloo, 2026. Advisor: Pascal Poupart.
   UWSpace

RESEARCH EXPERIENCE

Researcher, AI lab in stealth — Aug 2025 – Present

- Research on open-source computer-use agent models and on benchmark design for computer use
- Processing pipelines that turn large-scale human computer-use trajectories (screenshot–action pairs) into training and evaluation data

Student Researcher, Google — Jan – Dec 2025
Sponsored research collaboration with Pascal Poupart (Waterloo) and Suraj Kothawade (Google)

- Built Image-POSER, a reflective RL framework that casts image generation and editing as an MDP and learns to orchestrate a registry of pretrained text-to-image and image-to-image experts, supervised step by step by a vision–language model critic
- Investigated methods for improving the efficiency and inference-time cost of unified image generation models

Machine Learning Researcher, Vector Institute — Sep 2023 – Dec 2025

- Developed CRDA, a model-agnostic counterfactual augmentation method for tabular regression that reuses residuals as invariant noise under perturbations of statistically screened features; reduced MLP regressor MSE by 22.9% and XGBoost MSE by 6.4% on average across benchmark datasets
- Derived supporting theory, including a finite-sample filter consistency result and a causal robustness bound
- Ran distributed experiments on Alliance Canada HPC clusters

Research Assistant, University of Waterloo — Sep – Dec 2019
Advised by Prof. Mohammad Kohandel

- Researched interactions of cancer stem cells and their response to chemotherapy and immunotherapy
- Preprocessed and analyzed large experimental datasets to support the development of a statistical model

PRESENTATIONS

Counterfactual Residual Data Augmentation for Regression – Poster — Jul 2026
ICML 2026, Seoul

Counterfactual Residual Data Augmentation for Regression – Poster — Jul 2026
RIKEN–Vector–NAIRL Workshop, Seoul

Counterfactual Residual Data Augmentation for Regression – Talk — May 2026
Vector Institute Community Day, Toronto

Image-POSER: Reflective RL for Multi-Expert Image Generation – Talk — Oct 2025
Vector Institute Research Day, Toronto

HONOURS AND AWARDS

President's Graduate Scholarship, University of Waterloo — 2026
Ontario Graduate Scholarship — 2026
President's Research Award, University of Waterloo — 2020
President's Scholarship of Distinction, University of Waterloo — 2019
Dean's Honours List, University of Waterloo — 2018 – 2023

TEACHING

Teaching Assistant, CS 114: Principles of Computing for Science, University of Waterloo — Fall 2024

- Led tutorials and marked assignments and exams

Teaching Assistant, CS 231: Algorithmic Problem Solving, University of Waterloo — Spring 2024

- Led tutorials and marked assignments and exams

Teaching Assistant, Reinforcement Learning Bootcamp, Vector Institute — May 2024

- Taught foundational reinforcement learning methods and ran hands-on workshops for 10 industry partner teams
- Worked directly with teams to develop proof-of-concept RL projects, presented at the closing showcase

Guest Lecturer, CS 458: Computer Security and Privacy, University of Waterloo — Fall 2023

- Lectured on deep reinforcement learning for smart contract security: a DQN agent that interacts with Ethereum contracts to discover and exploit fund-leaking ("prodigal") vulnerabilities, with applications to pre-deployment testing

INDUSTRY EXPERIENCE

Software Development Engineer, Amazon Robotics — Jul 2023 – Feb 2025

- Built and operated production software for Amazon Robotics, the organization behind the robotic fulfillment systems deployed across Amazon's global warehouse network
- Designed distributed services for large-scale automated material handling, running continuously in live fulfillment centres

Software Engineer, co-operative terms

- BitGo (Jan – Apr 2023) · Amazon (May – Aug 2022) · StackAdapt (Jan – Apr 2022) · ApplyBoard (Sep 2020 – Apr 2021) · IBM (Jan – Apr 2020) · High Q (May – Aug 2019)

TECHNICAL SKILLS

Languages	Python, C, C++, Java, JavaScript/TypeScript, SQL
Machine learning	PyTorch, Hugging Face diffusers, Stable-Baselines3
Infrastructure	Slurm (Alliance Canada), Weights & Biases, AWS, Git